

Notes: Jin & Leslie (QJE, 2003)
**The Effect of Information on Product Quality: Evidence from Restaurant
Hygiene Grade Cards**

Contents

1	Big picture	2
2	Incremental contribution	2
3	Data, policy setting, and measurement	2
3.1	Inspection and revenue data	2
3.2	Policy regimes	2
3.3	Hospitalization data	2
4	Models and empirical specifications	3
4.1	Hygiene score regression	3
4.2	Revenue regression	3
4.3	Hospitalization regressions	3
4.4	Sorting versus real quality improvements	3
5	Identification and empirical design	3
6	Tables and figures	4
6.1	Figure I: Quartiles of Hygiene Quality Distribution over Time + Table I: Sample of Restaurants	4
6.2	Table II: Timing of Mandatory Grade Card Ordinance + Table III: Effects of Grade Cards and Disclosure Regulation on Hygiene Scores	4
6.3	Table IV: Revenue Effects + Figure II: Score Distributions	4
6.4	Tables V–VI: Hospital Admissions and Digestive Disorder Regressions	5
6.5	Table VII: Food-Related Illness Hospitalizations	6
7	Result map	6
8	Short summary	7
9	Conclusion	7
9.1	Core conclusion	7
9.2	Economic intuition	7
9.3	Policy implications	7
9.4	Limitations	7
9.5	Takeaway	7

1 Big picture

- **Question.** Does public disclosure of product-quality information cause firms to improve product quality?
- **Policy experiment.** Los Angeles County introduced standardized restaurant hygiene grade cards in January 1998.
- **Main outcomes.** The paper studies inspection scores, quarterly revenues, foodborne illness hospitalizations, and the distribution of inspection scores.
- **Main result.** Grade cards raise hygiene scores, make revenue responsive to hygiene grades, and reduce foodborne illness hospitalizations.
- **Economic takeaway.** Standardized consumer information disclosure can change both demand and firm quality choice.

2 Incremental contribution

- The paper provides a clean empirical test of disclosure policy using a sudden and unanticipated policy shock.
- It moves beyond price-advertising studies by studying disclosure of hard-to-observe product quality.
- It links disclosure to firm behavior, demand response, and real health outcomes.
- It separates score changes from true quality improvements by using hospitalization data and revenue-weighted health regressions.
- It provides suggestive evidence on mandatory versus voluntary disclosure under a standardized format.

3 Data, policy setting, and measurement

3.1 Inspection and revenue data

The paper uses Los Angeles County DHS inspection data from 1996–1998 matched to confidential sales-tax revenue data. The final revenue sample contains 13,544 restaurants.

Detailed interpretation. The inspection score is the quality measure, while revenue reveals whether consumers respond when the score becomes visible.

3.2 Policy regimes

After January 16, 1998 all restaurants received grade cards after inspection. Cities adopted mandatory posting at different times, creating three regimes: no standard card, voluntary card disclosure, and mandatory card disclosure.

Detailed interpretation. This generates both time-series and cross-sectional variation in information disclosure.

3.3 Hospitalization data

The paper uses California hospital admissions from 1995–1999, classified by food-related and nonfood-related digestive disorders at the ZIP-code level.

Detailed interpretation. Hospitalizations validate whether score improvements reflect true public-health improvements.

4 Models and empirical specifications

4.1 Hygiene score regression

$$h_{it} = \alpha_i + \beta_1 \text{Mandatory}_{it} + \beta_2 \text{Voluntary}_{it} + \gamma_1 \text{CriteriaII}_t + \gamma_2 \text{CriteriaIII}_t + \varepsilon_{it}.$$

Detailed interpretation. Restaurant fixed effects absorb time-invariant quality. Criteria dummies separate grade-card effects from scoring-rule changes.

4.2 Revenue regression

$$\log \text{Revenue}_{it} = \alpha_i + \tau_t + \text{Mandatory}_{it} + \text{Voluntary}_{it} + \text{Grade}_{it} + \text{Disclosure}_{it} \times \text{Grade}_{it} + \varepsilon_{it}.$$

Detailed interpretation. The interactions test whether consumers respond differently to A, B, C, and D grades after public disclosure.

4.3 Hospitalization regressions

$$\text{Admissions}_{zdt} = \text{fixed effects} + \text{FoodRelated}_d \times \text{Disclosure}_t + \text{controls} + \varepsilon_{zdt}.$$

Detailed interpretation. This compares food-related to nonfood digestive admissions before and after grade cards.

4.4 Sorting versus real quality improvements

$$\text{Admissions}_{it} = \beta_A A \text{Revenue}_{it} + \beta_B B \text{Revenue}_{it} + \beta_C C \text{Revenue}_{it} + \theta_1 \text{TotalRevenue}_{it} \times \text{GradeCards}_t + \theta_2 \text{Population}_i \times \text{GradeCards}_t + \varepsilon_{it}.$$

Detailed interpretation. If sorting fully explains illness reductions, revenue shares by hygiene tier should absorb the grade-card effect.

5 Identification and empirical design

The design uses a rapid and unanticipated policy shock, staggered city adoption of mandatory posting, and random inspection timing. Demand response is tested using revenue, while true quality improvement is tested using hospital admissions. The main threats are inspection score inflation, aggregation in hospitalization data, and anticipation in voluntary disclosure cities.

6 Tables and figures

6.1 Figure I: Quartiles of Hygiene Quality Distribution over Time + Table I: Sample of Restaurants

Read: Figure I shows a stable pre-policy hygiene distribution, a large July 1997 assessment shift, and continued improvement after grade cards. Table I shows that the final matched sample closely follows the full DHS sample.

Detailed interpretation. The pair establishes both the outcome trend and sample comparability.

Economic message. Grade-card disclosure coincides with a sharp improvement in the lower tail of hygiene scores.

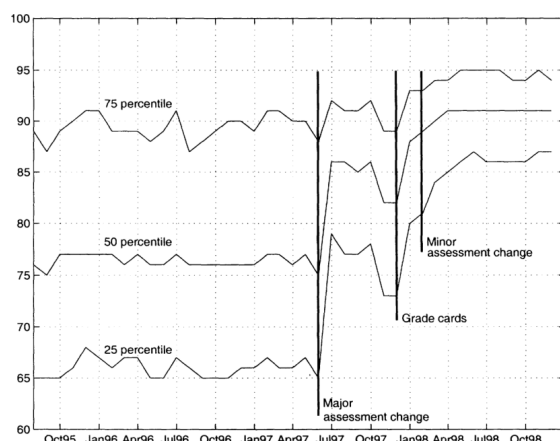


FIGURE I
Quartiles of Hygiene Quality Distribution over Time
Quartiles are computed based on all inspections in a given month. The assessment changes took place on July 1, 1997 and March 18, 1998. The grade card began introduction on January 16, 1998. Vertical lines for regime changes are located immediately prior to a change in order to emphasize subsequent impact on the hygiene distribution.

(a) Figure I

Quarter	DHS sample		Matched (SBE) sample		Our sample	
	No. restaurants	Ave. hygiene score	No. restaurants	Ave. hygiene score	No. restaurants	Ave. hygiene score
1996 Q1	19590	75.62	11917	75.02	9647	75.40
Q2	19774	75.37	12821	74.67	9862	74.97
Q3	20155	75.03	13079	74.45	10054	74.84
Q4	20389	75.27	13445	74.73	10159	74.95
1997 Q1	20665	75.81	13521	75.35	10198	75.75
Q2	20898	75.31	13555	74.56	10181	74.96
Q3	21558	83.99	13492	83.34	10128	83.63
Q4	21647	81.82	13364	81.04	10052	81.27
1998 Q1	21951	86.69	13328	86.10	10126	86.44
Q2	22147	90.26	13316	89.65	10238	89.93
Q3	22514	89.85	12867	89.27	10222	89.56
Q4	22652	90.30	12061	89.83	9883	90.01
Total	23921	81.22	17284	80.50	13544	80.86

In the DHS sample, a restaurant is defined as a physical location serving food other than prepackaged items. A restaurant is counted in a particular quarter if it is inspected during that quarter, or if it was inspected both before and after the quarter.
The matched SBE sample is derived from successful matches of the DHS inspection data with the SBE tax data.
Our sample includes all restaurants for which the DHS data are matched with the SBE data, and for which we observe continuously reported tax payments between the first and last payments of each restaurant.
Quarterly average hygiene scores are calculated using all inspections conducted on restaurants in the relevant sample during that quarter. Not all restaurants are inspected every quarter.

(b) Table I

6.2 Table II: Timing of Mandatory Grade Card Ordinance + Table III: Effects of Grade Cards and Disclosure Regulation on Hygiene Scores

Read: Table II documents staggered adoption and restaurant exposure to disclosure regimes. Table III shows mandatory and voluntary disclosure significantly raise scores.

Detailed interpretation. Table II supplies the treatment variation; Table III estimates the score effect.

Economic message. Visible disclosure increases firms' incentives to score well.

6.3 Table IV: Revenue Effects + Figure II: Score Distributions

Read: Table IV shows grades matter for revenue only after disclosure. Figure II shows bunching around 90 under grade-card regimes.

Panel A: Number of restaurants located in cities with or without the ordinance					
Ordinance not adopted			Ordinance adopted		
Quarter	Total restaurants	No. restaurants	% of restaurants	No. restaurants	% of restaurants
1998 Q1	10128	9626	95.06	500	4.94
Q2	10238	9806	97.18	6432	62.82
Q3	10222	2972	29.07	7250	71.93
Q4	9883	2009	20.33	7874	79.67

under Regime I: voluntary disclosure without standard format		under Regime II: voluntary disclosure with standard format		under Regime III: mandatory disclosure	
Quarter	% of restaurant days	Quarter	% of restaurant days	Quarter	% of restaurant days
1998 Q1	84.64	Q2	15.04	Q3	0.32
Q2	40.78	Q3	43.20	Q4	16.02
Q3	15.59	Q4	41.28		43.13
Q4	4.94		34.27		61.69

Every restaurant receives an official grade card following inspections conducted after January 16, 1998. However, the restaurant is only required to post the grade card if it is located in a city which has adopted the ordinance. Restaurants not yet inspected after January 16, 1998, fall under Regime I.

Table II

TABLE III
THE EFFECTS OF GRADE CARDS AND DISCLOSURE REGULATION ON HYGIENE SCORES

	Without fixed effects		With fixed effects	
	Coefficient	Std. error	Coefficient	Std. error
Mandatory disclosure	4.9432	1.1384***	4.3958	1.4046***
Voluntary disclosure	4.0585	0.3199***	3.2528	0.3550***
Inspection Criteria II	7.7192	0.9181***	8.0886	0.9907***
Inspection Criteria III	9.9838	1.2233***	10.4158	1.3542***
Observations	69,991			
No. restaurants	13,544			
R^2	0.3574		0.5874	

Regressions include city random effects (i.e., we cluster the standard errors by city with Huber-White standard errors).

In the regression without fixed effects, while not reported, we also include the following restaurant characteristics: food type, food style, seating capacity, liquor license dummy, DHS risk assessment, and city dummies. Stars denote significance levels: 99 percent confidence level (***), 95 percent confidence level (**), and 90 percent confidence level (*).

The voluntary disclosure dummy is for voluntary verifiable disclosure (i.e., grade cards are issued but posting is discretionary). The excluded dummy is for voluntary nonverifiable disclosure (i.e., prior to the introduction of grade cards).

Inspection Criteria II Dummy is for inspections carried out between July 1, 1997, and March 18, 1998. See text for further details.

Inspection Criteria III Dummy is for inspections carried after March 1998. See text for further details.

Table III

Detailed interpretation. Revenue results show consumer response; score bunching warns that scores alone may reflect threshold behavior.

Economic message. Disclosure creates demand rewards but can also create measurement incentives.

	Coefficient	Std. error
Mandatory disclosure	0.0569	0.0153***
Voluntary disclosure	0.0326	0.0149**
B-grade	-0.0074	0.0084
C-grade	0.0039	0.0074
D-grade	-0.0023	0.0057
Mandatory $\times$ B-grade	-0.0497	0.0151***
Mandatory $\times$ C-grade	-0.0670	0.0304**
Mandatory $\times$ D-grade	-0.0565	0.0437
Voluntary $\times$ B-grade	-0.0029	0.0128
Voluntary $\times$ C-grade	-0.0238	0.0216
Voluntary $\times$ D-grade	-0.0758	0.0469
Missing grade	-0.0001	0.0096
Observations	74,321	
R^2	0.9506	

The regression also includes a restaurant fixed effects, a full set of quarterly dummies and city-level random effects (i.e., we cluster the standard errors by city with Huber-White standard errors).

D-Grade is equivalent to any score below 70 (i.e., less than a C-grade). Missing Grade is for restaurants that have opened but have not yet been inspected.

Excluded dummy is for voluntary disclosure without a standard format. Interactions with A-grade are also excluded.

(a) Table IV

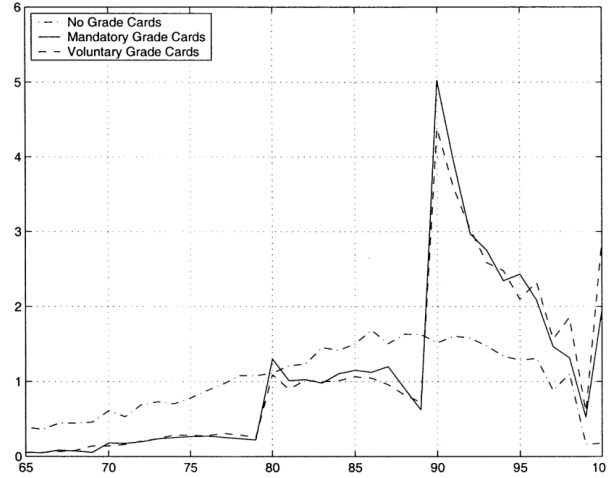


FIGURE II

Distributions of Hygiene Scores under Different Disclosure Regimes
The figure is no different from a histogram (or an unsmoothed nonparametric density). Units on the vertical axis are meaningless.

(b) Figure II

6.4 Tables V–VI: Hospital Admissions and Digestive Disorder Regressions

Read: Table V shows food-related admissions fall in Los Angeles after grade cards while comparison outcomes do not. Table VI estimates significant negative food-related disclosure interactions.

Detailed interpretation. These tables validate true health effects beyond inspection score changes.

Economic message. Disclosure reduces foodborne illness risk.

TABLE V
NUMBER OF HOSPITAL ADMISSIONS IN CALIFORNIA FOR DIGESTIVE DISORDERS

		Los Angeles County		California, except Los Angeles County	
		Food-related	Nonfood-related	Food-related	Nonfood-related
Year	Number	%	%	%	%
1995	401		54,412	607	128,949
1996	431	7.5%	56,692	675	131,623
1997	405	-6.0%	59,585	634	139,645
1998	351	-13.3%	61,305	654	145,261
1999	309	-12.0%	60,915	601	148,338

Data come from the *Office of Statewide Health Planning and Development* in California. We use version A of the data (i.e., for each patient we observe the month of admission and three-digit zip code). Digestive disorders are defined as any admission for which the major diagnostic category is 6 (MDC 6). We include only admissions where the patient is admitted from home as part of an unscheduled visit. An admission for a digestive disorder is counted as food-related if the principal diagnosis (using ICD-9-CM codes) is an illness that is transmitted via food in over 90 percent of occurrences. See text for further details.

(a) Table V

TABLE VI
THE EFFECTS OF GRADE CARDS ON ln (NO. HOSPITALIZATIONS
FOR DIGESTIVE DISORDERS)

	Coefficient	Std. error
Mandatory disclosure	0.0271	0.0246
Voluntary disclosure	0.0716	0.0238***
Food-related $\times$ mandatory disclosure	-0.2243	0.0426***
Food-related $\times$ voluntary disclosure	-0.2055	0.0350***
Observations	6,840	
R^2	0.9809	

Covariates not shown include fixed effects for food-related illnesses in each three-digit zip code, fixed effects for nonfood-related illnesses in each three-digit zip code, and year and month dummies. We also include three-digit zip code illness-type random-effects (i.e., we cluster the standard errors by three-digit zip code and illness-type with Huber-White standard errors). Stars denote significance levels: 99 percent confidence level (***), 95 percent confidence level (**), and 90 percent confidence level (*).

(b) Table VI

6.5 Table VII: Food-Related Illness Hospitalizations

Read: Table VII shows illness risk rises with lower hygiene-score revenue tiles and that grade-card effects remain after controlling for sorting.

Detailed interpretation. The hospitalization decline is not fully explained by consumers switching restaurants; actual hygiene quality appears to improve.

Economic message. The policy changes underlying firm quality, not only consumer allocation.

TABLE VII
THE EFFECTS OF GRADE CARDS ON THE NUMBER OF FOOD-RELATED
ILLNESS HOSPITALIZATIONS

	Coefficient	Std. error
A-tile revenue	-0.0146	0.0264
B-tile revenue	0.2892	0.0615***
C-tile revenue	1.1288	0.4367**
Total revenue $\times$ grade cards	0.0156	0.0140
Population $\times$ grade cards	-3.8327	1.0045***
Q1 $\times$ population	5.9537	1.0871***
Q2 $\times$ population	9.1979	0.7719***
Q3 $\times$ population	11.2465	1.3932***
Q4 $\times$ population	8.4846	1.1998***
Observations	191	
R^2	0.9156	

The regression also includes three-digit zip code random effects (i.e., we cluster the standard errors by three-digit zip code with Huber-White standard errors). Revenue variables are in units of 10^5 . Also, revenue is deflated using the BLS consumer price index for all urban consumers.

Table VII

7 Result map

- Grade cards raise inspection scores.
- Consumers respond to visible grades in revenue data.

- Score bunching suggests possible inspector or threshold behavior.
- Foodborne illness hospitalizations fall after grade cards.
- The health effect is not fully explained by consumer sorting.
- Standardized disclosure works even when initial posting is partly voluntary, though anticipation complicates interpretation.

8 Short summary

- The paper studies Los Angeles restaurant hygiene grade cards.
- It uses inspection, tax revenue, and hospitalization data.
- Grade cards increase scores, alter revenue patterns, and reduce food-related hospitalizations.
- Health regressions imply true hygiene improvements, not just grade inflation or sorting.

9 Conclusion

9.1 Core conclusion

Restaurant hygiene grade cards caused restaurants to improve hygiene quality and reduced foodborne illness hospitalizations.

9.2 Economic intuition

When hidden quality becomes visible, consumers reward high quality and punish low quality; firms then have stronger incentives to improve.

9.3 Policy implications

Standardized, visible, verifiable information disclosure can be a powerful regulatory tool in markets with asymmetric information.

9.4 Limitations

The paper does not observe prices and quantities separately; hospitalization data are aggregated; and score bunching indicates potential measurement distortions.

9.5 Takeaway

Information disclosure is not merely informational. It changes incentives and can improve real outcomes.